W12 Exercises

📌 Discuss with TAs or consult the TA session notes if you run into any difficulties!

Task 0

- Download the dataset `courage_log` import it as `log`, and enjoy the exercise!

Task 1: Courage the Cowardly Dog (S1)



💬 "We interrupt this program to bring you... Courage the Cowardly Dog Show! Starring Courage, the Cowardly Dog!
Abandoned as a pup, he was found by Muriel, who lives in the middle of Nowhere with her husband, Eustace Bagge.
[Eustace pulls on the monster mask] **"OOGA BOOGA BOOGA!"** *[Courage screams in terror]*
But creepy stuff happens in Nowhere. It's up to Courage to save his new home!"

- Courage the Cowardly Dog has a computer who can talk (or we can say he is the very first one dog using AI) Courage's computer has recorded every spooky encounter in Nowhere from Season 1! The log records the following:
  - `episode` : the episode identifier
  - `title` : the episode segment title
  - `monster_type` : the primary threat faced in the episode
  - `fear_level` : Courage's subjective terror level on a 1-10 scale
- Follow the instructions. Help Courage to analyze these terrifying events, and whether the type of threat faced is related to the fear-level he felt.

Step 1: Classify fear level

- Create a new column `high_fear` in `courage_log` that labels encounters as `TRUE` if `fear_level >= 8` and `FALSE` otherwise. Save the resulting dataframe as `log_classified`

Step 2: Grouped Summarization

- Compute a table showing the number of episode segments `n` and average fear rating (`mean_fear`) for each combination of `monster_type` and `high_fear` . Save the table as `summary_counts`
  - **Hint:** remember to `ungroup()` after your work!

Step 3: Visualize the Fear

- Create a stacked bar chart displaying the count of episode segments across `monster_type` , filled by `high_fear`
  - Hint: choose the correct plot function and use `geom_xxx(position = "stack")` to stack the count

Step 4: Contingency Table

- Use the data from `log_classified` . Create a contingency table containing `high_fear` x `monster_type`
- Then create a proportional table from the above table both by rows and by columns

Step 5: Chi-Squared Test

- Test the following hypothesis:
  - H_0: Threat severity (`high_fear`) is independent of `monster_type`
  - H_A: Threat severity (`high_fear`) is associated with `monster_type`
- Report your findings and interpret the results